



Nexcomm Systems · Wi-Fi HaLow

ALF Living Campus

Nexcomm HaLow + mooreVIEW — institutional / assisted living
Survey-ready campus ops before families complain or survey week arrives

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At a Glance

Contents

- [One-line pitch](#)
- [Reactive vs proactive](#)
- [Site architecture](#)
- [Node inventory \(reference campus\)](#)
 - [Phase roadmap](#)
- [Nexcomm device templates \(phase 2 drop-in\)](#)
- [Semantic tags → CMMS](#)
- [Commissioning checklist](#)
- [Proof metrics \(12-month pilot\)](#)
- [Gaps vs full Nexcomm MDU facility deck](#)
- [Related files](#)

ALF Living Campus — Nexcomm HaLow Story

Vertical: Institutional / assisted living (Living Campus) · Marketplace #10

Nexcomm reference: MDU facility diagram + HVAC split-system slide

MV project: `assisted-living-halow` · `assisted-living-pool-iot-link`

Pricing: Enterprise campus — see [Pricing Guide](#) (Living Campus tier)

One-line pitch

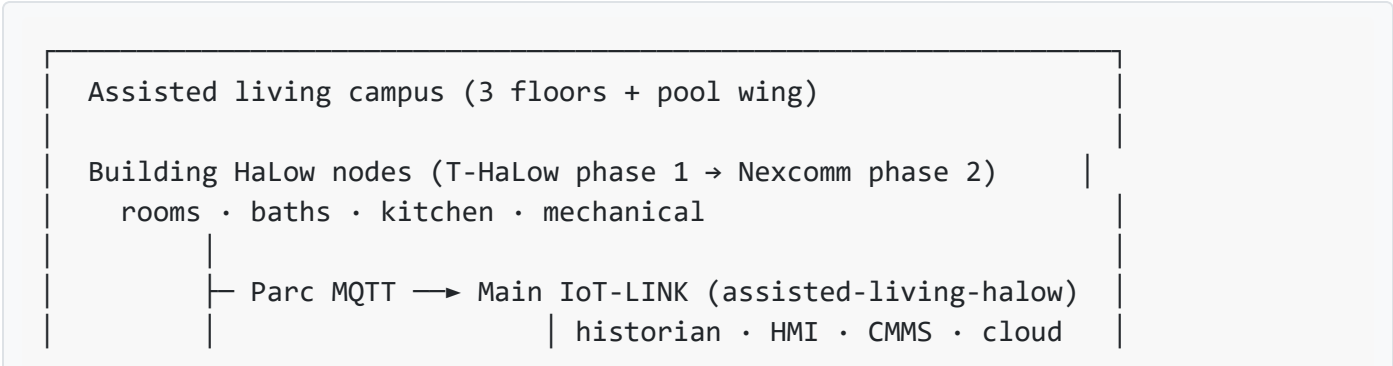
Nexcomm HaLow covers the **whole campus** — resident rooms, kitchen, mechanical, pool deck, and rooftop RTUs — on one long-range radio. mooreVIEW turns those tags into **survey-ready evidence and proactive CMMS** before families complain or survey week arrives.

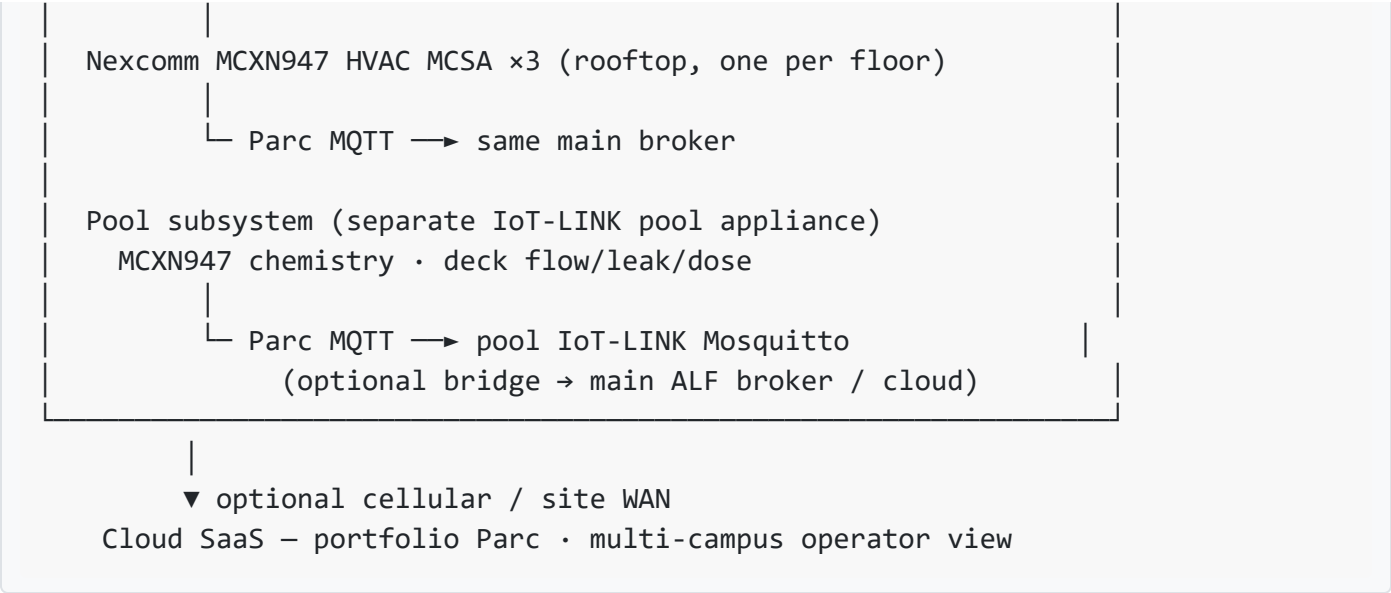
Reactive vs proactive

Reactive (today)	Proactive (mooreVIEW + Nexcomm)
Thermostat clouds, kitchen loggers, generator clipboards	One campus .est + Parc fleet
Leak found after water damage	Bath/kitchen HaLow leak tags → WO in minutes
RTU fails when residents complain	Rooftop MCSA on compressor/fan → PdM WO days ahead
Survey prep = fire drill	Continuous evidence export
Pool chemistry on a clipboard	MCXN947 probe + deck I/O over HaLow to pool IoT-Link

Scope: Facilities / MEP only — not clinical, nurse call, EHR, or PHI.

Site architecture





Gateway roles

Role	Hardware	Broker
Main campus	Compulab IoT-Link (mv_alf_main_01)	mqtt://127.0.0.1:1883 on appliance
Pool plant	IoT-Link pool (mv_alf_pool_01)	Local Mosquitto — HaLow anchor for pool equipment room
HaLow AP	Nexcomm Nexus AP or site 802.11ah AP	Routes HaLow LAN to broker IP
Backhaul	Cellular Opta gateway or site Ethernet	Cloud remote enrollment

Node inventory (reference campus)

Maps Nexcomm **MDU facility** (building + grounds) and **HVAC monitoring** slides to deployed nodes.

Zone	Device	Platform	Template / firmware	Signals
Mechanical room	thalow_mech_01	LilyGO T-HaLow	ALF #1	Utility flow / meter pulse
Kitchen refer/freezer	thalow_kitchen_01	T-HaLow	ALF #5	Freezer + reefer temp
Kitchen six sinks	thalow_kitchen_sinks	T-HaLow	ALF #6	Leak x6
Resident room (xN)	thalow_rm###_room	T-HaLow	ALF #2	Comfort / pan leak

Resident bath (×N)	thalow_rm###_bath	T-HaLow	ALF #3	Toilet / sink leak
Common bath	thalow_comm_*	T-HaLow	ALF #3	Toilet leak
Floor 1–3 rooftop	hvac_mcsa_fl1..3	Nexcomm MCXN947	mcsa-hvac	T1–T4, COMP_FLT, FAN_FLT, WR_DETECT
Therapy pool chemistry	pool_sensor_01	MCXN947 pool	mcxn947-pool-sensor	pH, ORP, cond, water temp
Pool deck	thalow_pool_deck	T-HaLow	custom	Flow, leak, dose relays

Full CSV: `st/fixtures/assisted_living_halow_nodes.csv`

Manifest: `st/fixtures/assisted_living_halow.json`

Phase roadmap

Phase	Field radio	Leak / env	HVAC rooftop
1 (now)	LilyGO T-HaLow	Parc templates #1–#6	Nexcomm MCXN947 MCSA
2	Nexcomm HaLow leak / Halo / BME688	nexcomm/halow, nexcomm/halo, nexcomm/env MQTT	Same MCSA firmware

Phase flag in manifest: `"field": "t-halow", "future": "nexcomm"` .

Nexcomm device templates (phase 2 drop-in)

Template ID	Use on campus
nexcomm_halow_leak_mqtt	Bath/kitchen leak, flow, totalizer (6 ch)
nexcomm_halo_mqtt	Room temp/RH base + up to 10 expansions
nexcomm_bme688_env_mqtt	IAQ — temp, RH, VOC, CO ₂ eq
mcxn947_hvac_mcsa	Rooftop split/RTU compressor + fan MCSA

MQTT topic roots: `nexcomm/halow/<id>/telemetry` , `nexcomm/halo/<id>/telemetry` , `nexcomm/env/<id>/telemetry` .

HVAC MCSA uses native **Parc** `mooreview/v1/<deviceId>/telemetry` .

Semantic tags → CMMS

Representative bindings (see `scripts/assisted-living-halow/halow-semantic-map.js`):

Campus tag	Source	Proactive trigger
RM101_TOILET_LEAK	thalow_rm101_bath · FLOW_TOILET	Critical WO — water damage
KITCH_REEFER_TEMP	thalow_kitchen_01 · TEMP_REF	Kitchen cold drift → WO before food loss
FL1_COND_COMP_FLT	hvac_mcsa_f11 · COMP_FLT	PdM / MCSA → planned RTU service
POOL_PH	pool_sensor_01 · PH_AI	Chemistry band → pool WO
MECH_METER_KWH	thalow_mech_01	Utility anomaly (optional)

HVAC condenser bindings: `scripts/assisted-living-halow/hvac-halow-bindings.js`
(`FLn_COND_*` tags).

Workflow: tag alarm or PdM forecast → CMMS work order (source: `alarm` | `pdm` | `pm`) → technician closes with service history → next forecast learns from repair.

Commissioning checklist

1. **Base project** — `node scripts/assisted-living/generate-est.js` (full campus `.est` + HMI).
2. **HaLow overlay** — `node scripts/assisted-living-halow/generate-artifacts.js` → `assisted-living-halow.est.json`, pool est, env examples.
3. **Main IoT-Link** — copy `deploy/iot-link/.env.assisted-living-halow.example`; open `assisted-living-halow.est.json`.
4. **Pool IoT-Link** — copy `.env.assisted-living-pool.example`; RS-485 pump bus; `30_pool_controller.st`.
5. **HaLow AP** — broker IP reachable from HaLow LAN (often AP gateway, e.g. `10.10.10.1`).
6. **T-HaLow nodes** — provisioning AP `MooreVIEW-T-HaLow` → `/setup` → broker, deviceId, sensor template.
7. **Nexcomm HVAC** — flash `mcsa-hvac`; set `MV_DEVICE_ID=hvac_mcsa_f1n`, broker on HaLow LAN.
8. **Sync tags** — Drivers → Sync; verify Parc hub sees all `deviceId` s.
9. **Alarm drill** — leak + kitchen temp + COMP_FLT → CMMS WO with trend context.
10. **Cloud** — enroll gateway; portfolio operator view for multi-campus rollouts.

Training: **M14 — Parc edge peers (T-HaLow & cellular)** · `halow-xiao-sta/README.md`.

Proof metrics (12-month pilot)

- Evidence pack used **before** survey window (not the week before).
 - $\geq 60\%$ critical WOs opened from **leading indicators**, not resident complaints.
 - Rooftop MCSA case: compressor/fan fault $\rightarrow$ **planned** WO ≥ 7 days before hard failure.
 - 2–3 campuses on one Parc tenant for regional operator.
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Gaps vs full Nexcomm MDU facility deck

Not in v1 ALF story (future vertical expansion):

- Access control
- Irrigation control
- Outdoor lighting monitor/control

These remain Nexcomm-native or custom I/O; MV core value stays MEP + leaks + cold + HVAC + pool.

Related files

Path	Purpose
scripts/assisted-living-halow/alf-halow-data.js	Node manifest source
scripts/assisted-living-halow/generate-artifacts.js	Est + fixture generator
data/projects/assisted-living-halow.est.json	Generated campus project
deploy/iot-link/.env.assisted-living-halow.example	Main appliance env
test/assistedLivingHalow.test.js	Manifest + HVAC node tests
06-institutional-summary.md	Marketplace 1-pager